Arista Access Node Optical Transport Failure Resolution Procedure

Node: KOR-ACC-66 | **Vendor:** Arista | **Severity:** Critical | **Alarm Code:** ALARM_7163

Description

The Arista access node, situated at the Koramangala Datacenter with the node ID KOR-ACC-66, is susceptible to optical transport failures due to the inherent complexities of optical signal transmission. This failure domain, denoted by alarm code ALARM_7163, is precipitated by a multitude of factors including SFP/XFP degradation, fiber cuts, and Rx/Tx power loss. The Arista access node, operating as an aggregation point for numerous downstream devices, relies on the stability of its optical interfaces to maintain uninterrupted data transmission. The optical transport failure can be attributed to the physical properties of light transmission, where signal attenuation, dispersion, and noise can compromise the integrity of the data being transmitted. Furthermore, the Arista EOS (Extensible Operating System) relies on a complex interplay of protocols, including the Link Layer Discovery Protocol (LLDP) and the Optical Transport Network (OTN) protocol, to ensure seamless data transmission. Therefore, a comprehensive understanding of these protocols and the underlying hardware mechanics is essential to diagnose and rectify the optical transport failure.

Pre-requisite Checks

- Dispatch remote hands to the Koramangala Datacenter to perform physical checks

- Verify the status of redundant links to ensure no single point of failure

- Check the environmental conditions, such as temperature and humidity, to ensure they are within the specified ranges for the Arista equipment

- Confirm the Arista EOS version and ensure it is compatible with the latest security patches and updates

Potential Root Causes

Cause	Probability
SFP/XFP degradation	High
Fiber cuts	Medium
Rx/Tx power loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show interfaces transceiver details, Expected Output:
Detailed information about the transceiver, including the wavelength,
power levels, and signal-to-noise ratio, Description: This command
isolates issues related to the optical transceivers, providing
insights into the signal quality and power levels, which can indicate
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SFP/XFP degradation or fiber-related issues.

Command: show interfaces optical, Expected Output: Information about the optical interfaces, including the current optical power levels and any error messages, Description: This command provides real-time data about the optical interfaces, enabling the identification of issues related to Rx/Tx power loss or fiber cuts.
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Resolution Steps

Step 1: Step: Replace faulty SFP/XFP, Command: replace transceiver / with , Description: Replacing the faulty SFP/XFP with a new one can resolve issues related to SFP/XFP degradation. Ensure the new transceiver is compatible with the Arista equipment and the specific interface.

Step 2: Step: Repair or replace fiber cuts, Command: no shutdown , Description: If a fiber cut is detected, the affected fiber must be repaired or replaced. Once the fiber is restored, the interface can be re-enabled using the 'no shutdown' command.

Post-Check Verification

```
Command: show interfaces transceiver details, Expected Output: Updated information about the transceiver, indicating improved signal quality and power levels, Description: After replacing the faulty SFP/XFP or repairing the fiber cut, this command verifies the improved signal quality and power levels, ensuring the optical transport failure has been resolved.

Command: show interfaces optical, Expected Output: Updated information about the optical interfaces, indicating stable optical power levels and no error messages, Description: This command confirms the stability of the optical interfaces, ensuring the resolution steps have successfully addressed the optical transport failure.
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Roll-back Procedures

- Step: Revert to previous SFP/XFP, Command: replace transceiver / with , Description: If the replacement SFP/XFP does not resolve the issue, revert to the previous transceiver to maintain the original configuration.

- Step: Disable and re-enable the interface, Command: shutdown ; no shutdown , Description: If the fiber repair or replacement does not resolve the issue, disable and re-enable the interface to reset the optical connection and ensure the interface is functioning correctly.